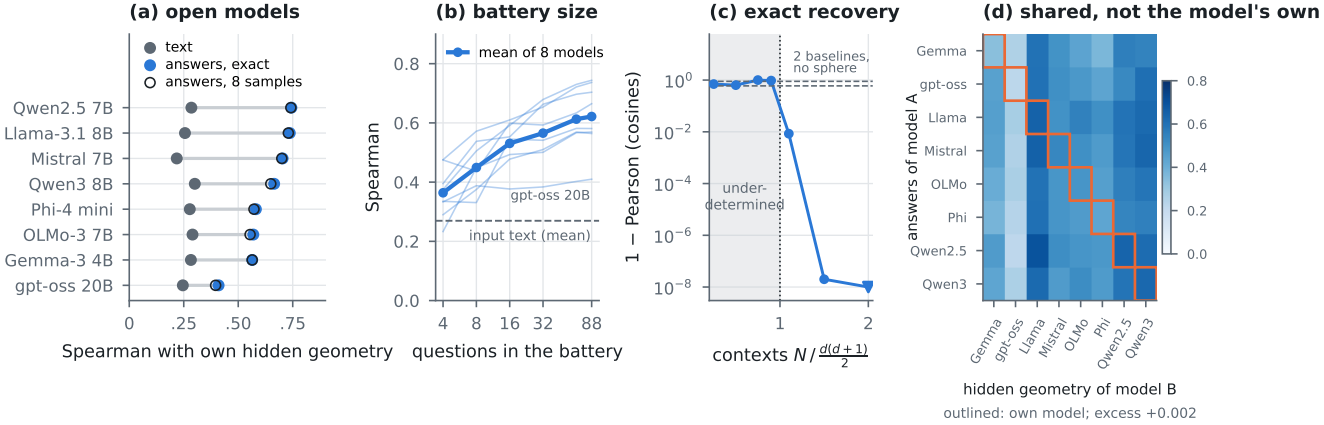
(a) open models
text
answers, exact
answers, 8 samples
Qwen2.5 7B
Llama-3.1 8B
Mistral 7B
Qwen3 8B
Phi-4 mini
OLMo-3 7B
Gemma-3 4B
gpt-oss 20B
0
.25
.50
.75
Spearman with own hidden geometry
(b) battery size
mean of 8 models
Spearman
0.8
0.6
0.4
0.2
0.0
gpt-oss 20B
input text (mean)
4
8
16
32
88
questions in the battery
(c) exact recovery
1 − Pearson (cosines)
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
2 baselines, no sphere
under-determined
1
2
contexts $N / \frac{d(d+1)}{2}$
(d) shared, not the model's own
answers of model A
Gemma
gpt-oss
Llama
Mistral
OLMo
Phi
Qwen2.5
Qwen3
0.8
0.6
0.4
0.2
0.0
hidden geometry of model B
outlined: own model; excess +0.002